

List of Experiments:

Computer Networks Experiments:

1. Implement the data link layer framing methods such as character, character-stuffing and bit stuffing.
2. Write a program to compute CRC code for the polynomials CRC-12, CRC-16 and CRC CCIP
3. Develop a simple data link layer that performs the flow control using the sliding window protocol, and loss recovery using the Go-Back-N mechanism.
4. Implement Dijkstra's algorithm to compute the shortest path through a network
5. Take an example subnet of hosts and obtain a broadcast tree for the subnet.
6. Implement distance vector routing algorithm for obtaining routing tables at each node.
7. Implement data encryption and data decryption
8. Write a program for congestion control using Leaky bucket algorithm.
9. Write a program for frame sorting technique used in buffers.
10. Wireshark
 - i. Packet Capture Using Wire shark
 - ii. Starting Wire shark
 - iii. Viewing Captured Traffic
 - iv. Analysis and Statistics & Filters.
11. How to run Nmap scan
12. Operating System Detection using Nmap
13. Do the following using NS2 Simulator
 - i. NS2 Simulator-Introduction
 - ii. Simulate to Find the Number of Packets Dropped
 - iii. Simulate to Find the Number of Packets Dropped by TCP/UDP
 - iv. Simulate to Find the Number of Packets Dropped due to Congestion
 - v. Simulate to Compare Data Rate & Throughput.
 - vi. Simulate to Plot Congestion for Different Source/Destination
 - vii. Simulate to Determine the Performance with respect to Transmission of Packets

Introduction to Lab:

1. COMPUTER NETWORKS

Definition:

A computer network is a group of computer systems and computing hardware devices that are linked together through communication channels to facilitate communication and resource-sharing among a wide range of users.



Applications:

Major applications of computer networks are:

1. Business Applications
2. Home Applications
3. Mobile Computers

1. Business Applications:

- Use of computers in business field to monitor production, inventories, to make payments.
- Resource sharing is the important purpose of using computer networks.

2. Home Applications:

Use of computers in home is widespread. Popular uses of computers in home are:

- Internet access.
- Personal Communication.
- Entertainment.
- Electronic Commerce.

3. Mobile Computers:

Many users use desktop computers at office and want to be connected to the office network while traveling. This is possible by wireless networks. Wireless networks are used in:

- Taxis, delivery vehicles and other mobile vehicles for keeping contacts with their office.
- Geographical Information Systems (GIS).

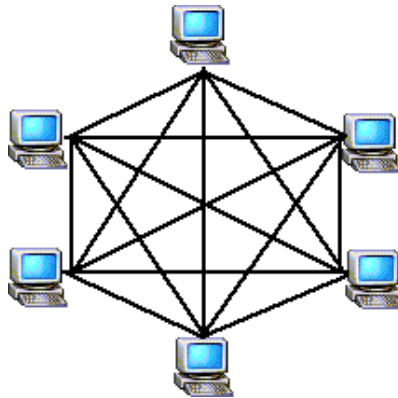
- Military applications.
- Air ports.
- Banking
- Weather reporting.

2. NETWORK TOPOLOGIES

- A network topology is the geometric representation of the all the links and linking devices to one another
- Five types of topologies are commonly used in the network. They are bus, star, ring, mesh and hybrid topology.

Mesh Topology:

- In a mesh network topology, each of the network nodes are interconnected with one another.
- Every node not only sends its own signals but also relays data from other nodes. It is commonly used in wireless networks.
- Flooding or routing technique is used in mesh topology.



Mesh Topology Diagram

Advantages of Mesh topology:

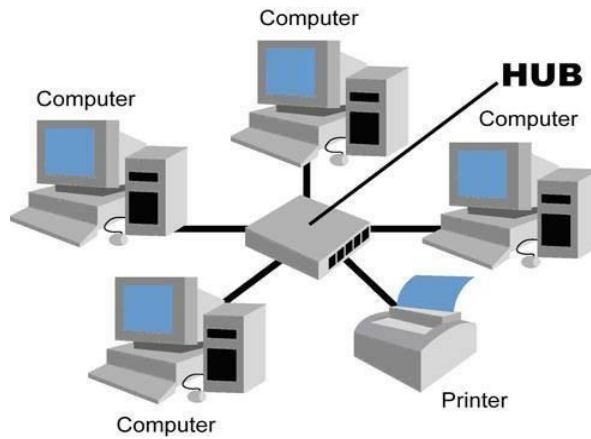
1. Troubleshooting is easy.
2. Isolation of network failures is easy.

Disadvantages of Mesh topology:

1. Difficulty of installation.
2. Costly because of maintaining redundant links.
3. Difficulty of reconfiguration.

Star Topology:

- In Star topology, all the components of network are connected to the central device called "hub".
- In this topology all the workstations are connected to central device with a point- to- point connection.



Star Topology Diagram

Advantages of Star Topology:

1. In star topology new nodes can be added or removed easily without affecting rest of the network
2. Troubleshooting is easy.

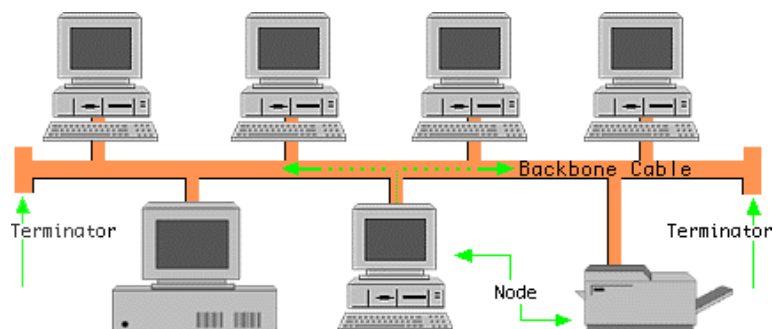
Disadvantages of Star Topology:

1. If the central hub fails, the whole network fails to operate.
2. Each device requires its own cable segment.

Bus

Topology:

- Bus Topology is the simplest of [network topologies](#). In this all the nodes are connected to the single cable, by the help of interface connectors.
- This central cable is the backbone of the network.



Bus topology diagram

Advantages Bus Topology:

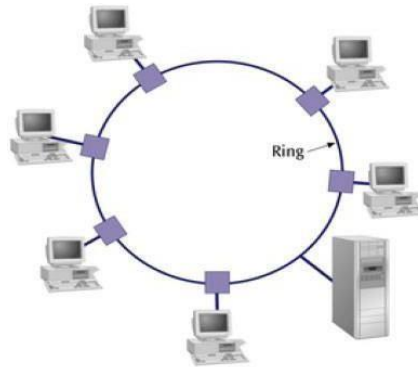
1. Easy to use and easy to install.
2. Bus topology costs very less.
3. Linear Bus network is mostly used in small networks.

Disadvantages Bus Topology:

1. It is difficult to detect and troubleshoot fault at individual station.
2. Maintenance costs can go higher with time.
3. Efficiency of Bus network reduces as the number of devices increases.

Ring Topology:

- In Ring Topology, all the nodes are connected to each-other in such a way that they make a closed loop.
- Data travels around the network, in one direction.



Advantages of Ring Topology:

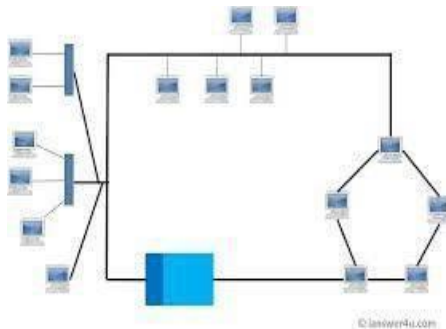
1. Cable failures are easily found.
2. Additional components do not affect the performance of network.

Disadvantages of Ring Topology:

1. It is difficult to troubleshoot a ring network.
2. Cost of cable is more in ring network.

Hybrid Topology:

- A network can be hybrid. Consider a main star topology with each branch connecting several stations in a bus topology.



Advantages of Hybrid Topology:

1. Unlike other networks, fault detection and troubleshooting is easy in this type of topology.
2. It is the combination of two or more topologies.

Disadvantages of Hybrid Topology:

1. One of the biggest drawbacks of hybrid topology is its design.
2. The hubs used to connect two distinct networks, are very expensive.

3. NETWORK TYPES

LAN - Local Area Network:

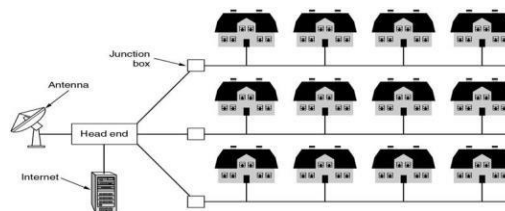
- A [LAN](#) connects network devices over a relatively short distance.
- The LAN transmission capacity is more than 1Mbps.
- The geographical coverage of LAN's is limited to areas less than 5 square kilometers.
- A networked office building, school, or home usually contains a single LAN.



MAN-Metropolitan Area Network:

- A **MAN** (*Metropolitan Area Networks*) connects multiple geographically nearby LANs to one another over an area of up to a few kilometers at high speeds.
- Thus, a MAN lets two remote nodes communicate as if they were part of the same local area network.
- A MAN is made from [switches](#) or routers connected to one another with high-speed links usually fiber optic cables.

Metropolitan Area Networks

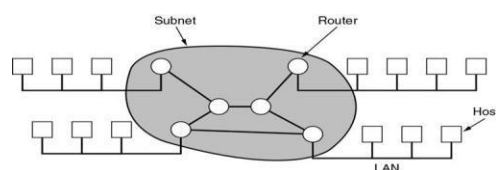


A metropolitan area network based on cable TV.

WAN-Wide Area Network:

A **WAN** spans a large geographic area, such as a state, province or country. WANs often connect multiple smaller networks, such as local area networks (LAN) or metro area networks (MAN). The world's most popular WAN is the Internet.

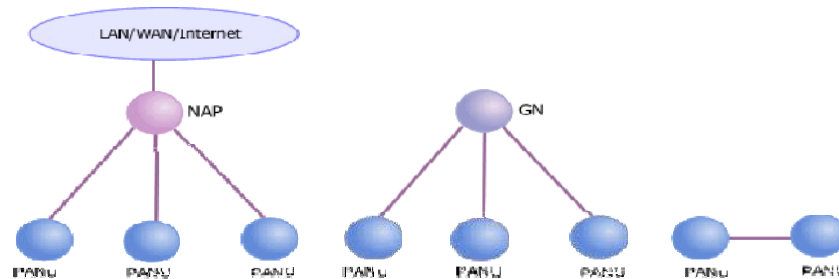
Wide Area Networks



Relation between hosts on LANs and the subnet.

PAN-Personal Area Network:

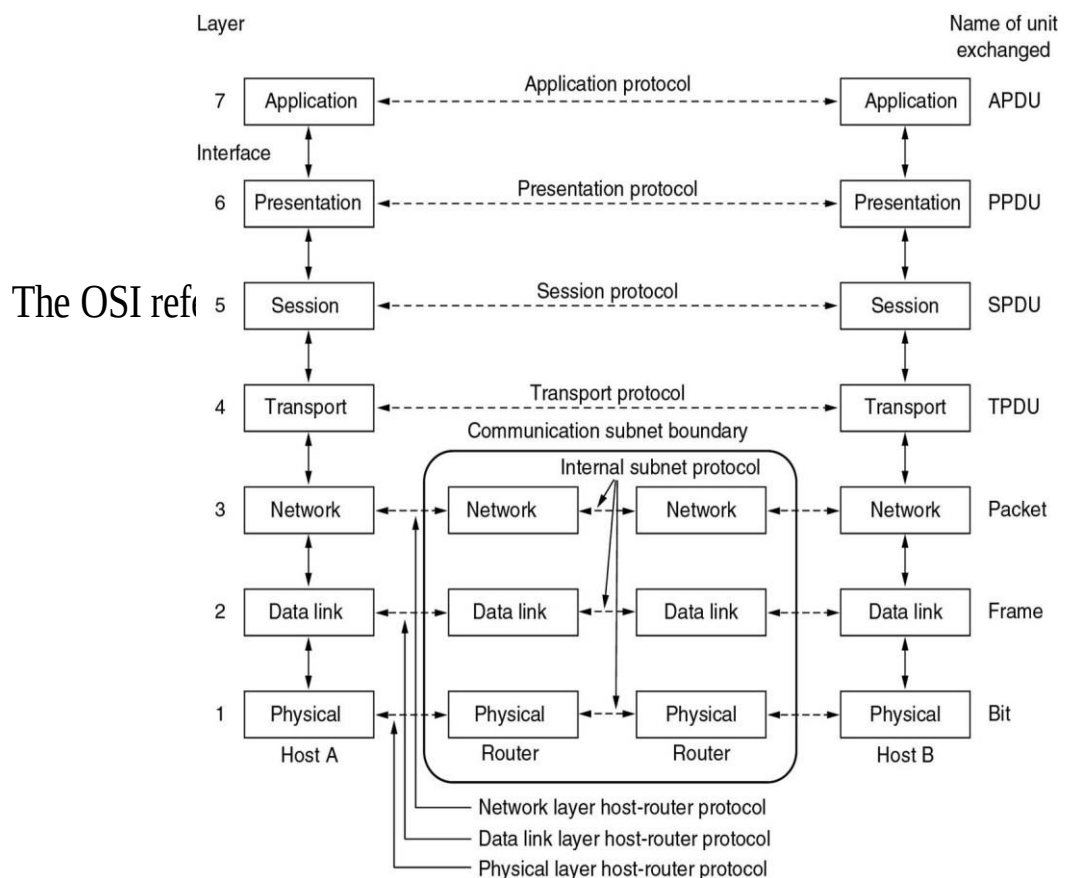
A personal area network (PAN) is the interconnection of information technology devices within the range of an individual person, typically within a range of 10 meters.



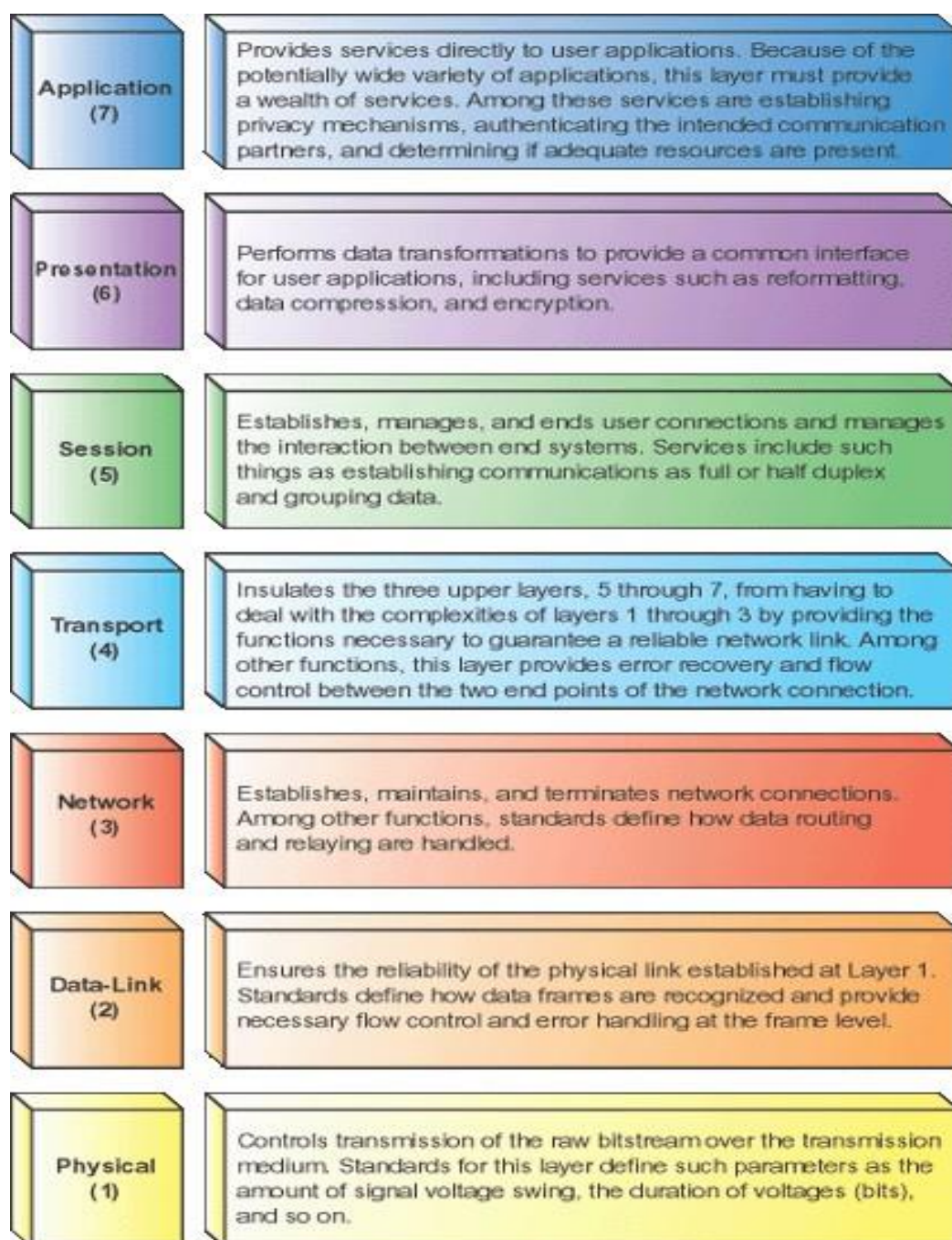
4. OSI / ISO MODEL

- The International Standards Organization for Standardization (ISO) developed the Open System Interconnection (OSI) reference model. OSI model is most widely used model for networking.
- An Open System is a set of protocols that allows any two different systems to communicate.

Reference Models

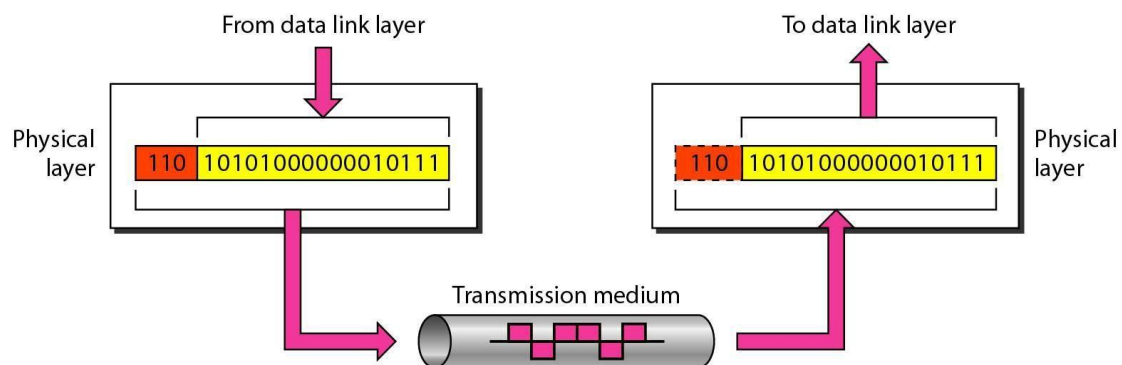


Seven layers of OSI model:



PHYSICAL LAYER:

- Physical layer is the lowest layer of the OSI model.
- The Physical Layer defines the electrical and physical specifications for devices.



Functions of Physical layer:

1. Physical characteristics of interfaces and media:

- The physical layer defines the characteristics of the interface between the devices and the transmission medium.
- It also defines the type of transmission medium.

2. Representation of bits:

- The physical layer data consists of a stream of bits (sequence of 0's or 1's) with no interpretation.
- To be transmitted, bits must be encoded into signals electrical or optical. The physical layer defines the type of encoding.

3. Data rate:

The transmission rate the number of bits sent each second is also defined by the physical layer.

4. Synchronization of bits:

The sender and receiver not only must use the same bit rate but also must be synchronized at the bit level.

5. Line configuration:

- The Physical layer is concerned with the connection devices to the media.
- In a point-to-point configuration, two devices are connected through a dedicated link. In a multipoint configuration, a link is shared among several devices.

6. Physical topology:

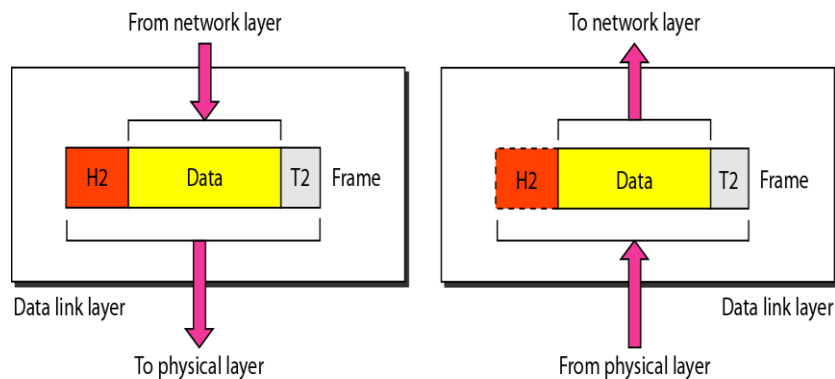
- The physical topology defines how devices are connected to make a network.
- Devices can be connected by using a mesh topology, a star topology, a ring topology, or a hybrid topology.

7. Transmission mode:

- The physical layer also defines the direction of transmission between two devices: simplex, half-duplex, Full-duplex.
- In simplex mode, only one device can send, the other receives. The simplex mode is a one-way communication.
- In the half-duplex mode, two devices can send and receive, but not at the same time.
- In full-duplex mode, two devices can send and receive at the same time.

DATA LINK LAYER:

- The data link layer is responsible for transmitting the frames from one node to the next.
- It transforms the physical layer to a reliable link making it an error free link to upper layer.



Functions of Data Link Layer:

1. Framing:

- The frames received from network layer are divided into manageable data units called frames.

2. Physical addressing:

- When frames are to be distributed to different systems on the network, the data link layer adds a header to the frame to define the sender and receiver of the frame.
- If the frame is intended for a system outside the sender's to the next one.

3. Flow control:

- If the rate at which the data are absorbed by the receiver is less than the rate at which data are produced in the sender, the data link layer imposes a flow control mechanism to avoid overwhelming the receiver.

4. Error control:

- The data link layer adds reliability to the physical layer by adding mechanisms to detect and retransmit damaged or lost frames. It also uses a mechanism to recognize duplication frames.

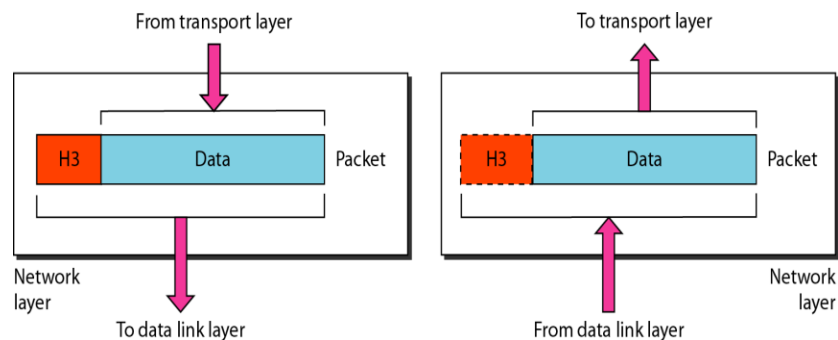
- Error control is normally achieved through a trailer added to the end of the frame.

5. Access control:

- When two or more devices are connected to the same link, data link layer protocols are necessary to determine which device has control over the link at any given time.

NETWORK LAYER:

The network layer is responsible for the delivery of packets from the source to destination across multiple networks.



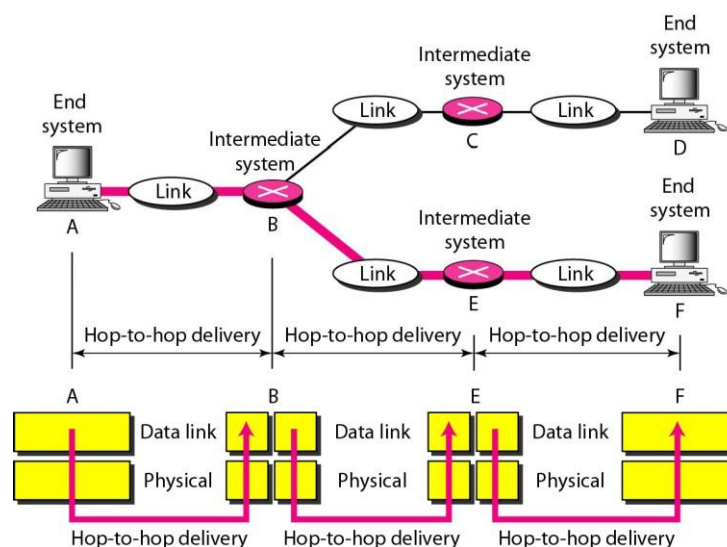
Functions of Network layer:

1. Logical addressing:

- The physical addressing implemented by the data link layer handles the addressing problem locally.
- If a packet passes the network boundary, we need another addressing system to help distinguish the source and destination systems.

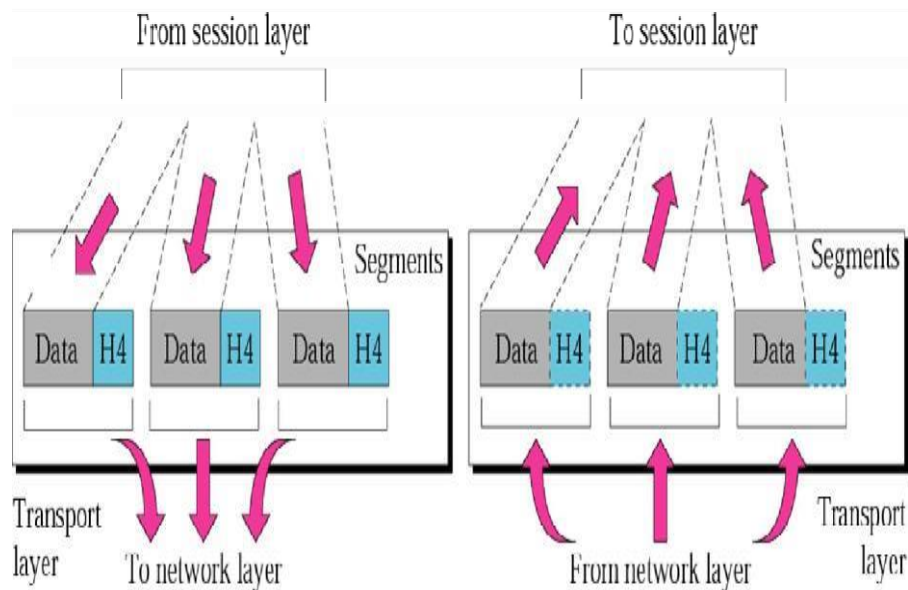
2. Routing:

- When independent networks or links are connected to create internet works or a large network, the connecting devices called routers or switches route or switch the packets to their final destination.
- One of the functions of the network, layer is to provide this mechanism



TRANSPORT LAYER:

The transport layer is responsible for process-to-process delivery of the entire message. A process is an application program running on a host.



Functions of transport layer:

1. Service-point addressing:

- Computer performs several operations simultaneously. Process-to-process delivery means specific process of one computer must be delivered to specific process on other computer. The transport layer header therefore includes port address.
- Network layer delivers packet to the desired computer and transport layer, gets message to the correct process on that computer.

2. Segmentation and reassembly:

- A message is divided into segments, each segment contains a sequence number which enables transport layer to reassemble at destination.

3. Connection control:

- Transport layer performs connectionless or connection oriented services with the destination machine.

4. Flow control:

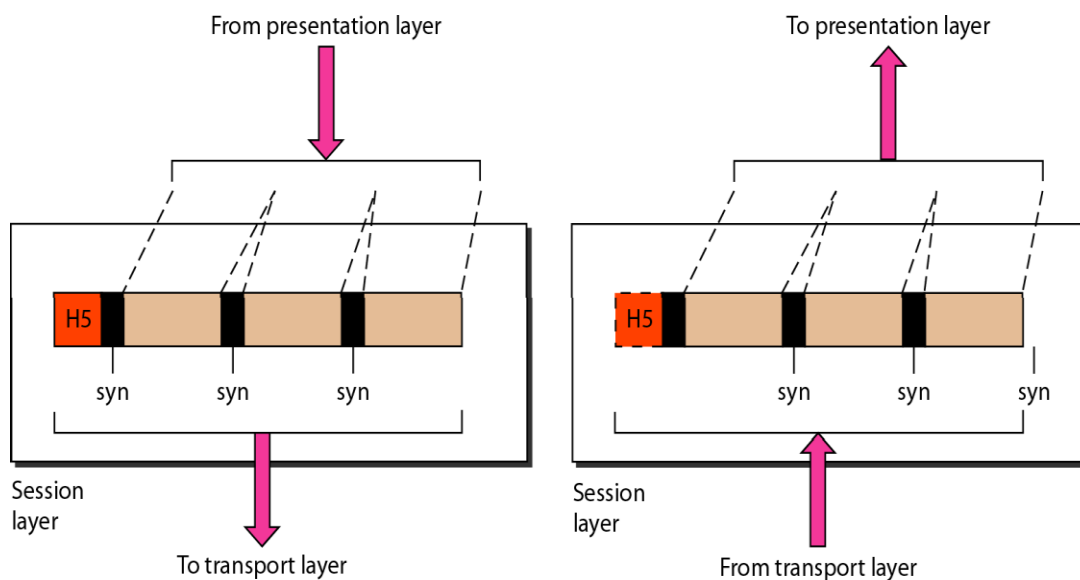
- Transport layer performs end-to-end flow control while data link layer performs it across the link.

5. Error control:

- Error control at this layer is performed on end-to-end basis rather than across the link. The transport layer ensures error free transmission.

SESSION LAYER:

The session layer is network dialog controller that is it establishes and synchronizes the interaction between communication systems.



Functions of session layer:

1. Dialog control:

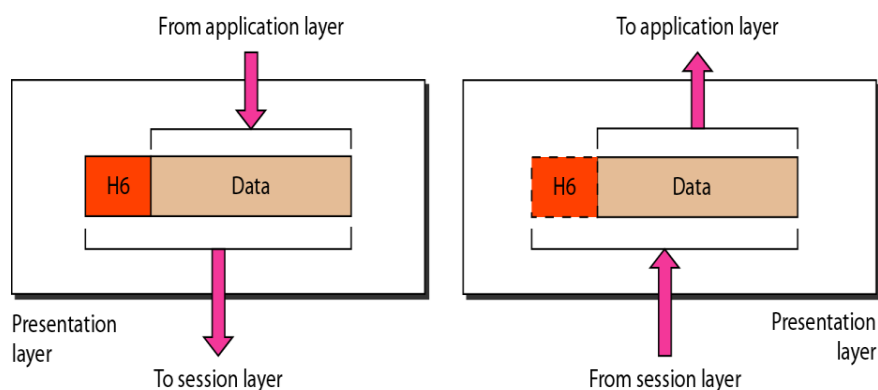
- Communication between two processes takes place in either half duplex or full duplex mode. The session layer manages dialog control for this communication.

2. Synchronization :

- Session layer adds synchronization points into stream of data.

PRESENTATION LAYER:

The presentation layer deals with syntax and semantics of the information being exchanged.



Functions of Presentation Layer:

1. Translation:

- Different computers use different encoding systems. The presentation layer maintains interoperability between the two encoding systems.

2. Encryption:

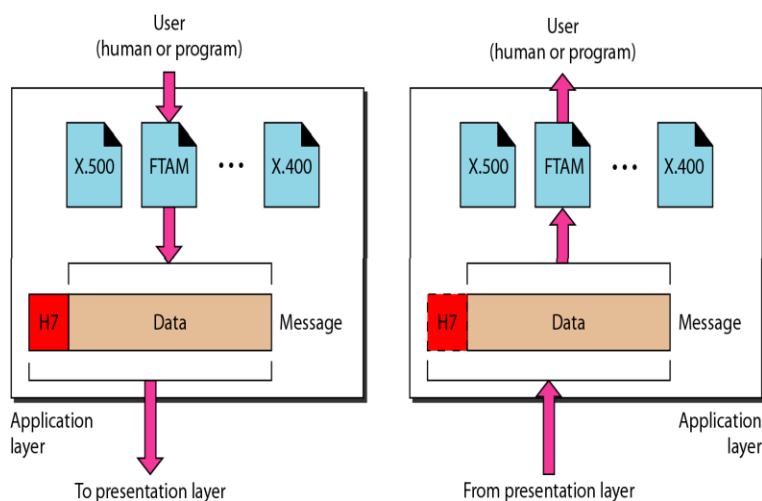
- Encryption is transforming sender information to other form to ensure privacy while transmission. Decryption is a reverse process.

3. Compression:

- Compression is a technique of reducing number of bits required to represent the data.

APPLICATION LAYER:

Application layer is responsible for accessing the network by user. It provides user interfaces and other supporting services such as e-mail, remote, file access, file transfer, sharing database, message handling, and directory services.



Functions of Application Layer:

1. Network Virtual Terminal:

- It is a software version of physical terminal that allows a user to log onto a remote host.

2. File Transfer, Access and Management (FTAM):

- FTAM allows user to access files in remote hosts, to retrieve files and to manage files in remote computer.

3. Mail Services:

- E-mail forwarding, storage are the services under this category.

4. Directory Services:

- Directory services include access for global information and distributed database.

Computer Networks Programs with Viva questions

PROGRAM-1

Implement the data link layer framing methods such as character, character-stuffing and bit stuffing.

DESCRIPTION:

- To prevent *data* being interpreted as *control* information. For example, many frame-based protocols, such as X.25, signal the beginning and end of a frame with six consecutive 1 bits. Therefore, if the actual data being transmitted has six 1 bits in a row, a zero is inserted after the first 5 so that the data is not interpreted as a frame delimiter. Of course, on the receiving end, the stuffed bits must be discarded.
- For protocols that require a fixed-size frame, bits are sometimes inserted to make the frame size equal to this set size.
- For protocols that required a continuous stream of data, zero bits are sometimes inserted to ensure that the stream is not broken.

1.1 CHARACTER COUNT

DESCRIPTION:

- It is used to count the number characters in the given text and
- To divide the total text in to various parts according to given frame length.

SOURCE CODE:

```
#include <stdio.h>
#include <string.h>

int main() {
    char data[100];
    int frame_size;
    int i, j, k;
    int current_pos = 0;

    printf("Enter the data string: ");
    scanf("%s", data);

    printf("Enter the desired frame size: ");
    scanf("%d", &frame_size);
    printf("\n The Total number of chars are :%d ",strlen(data));

    printf("\nTransmitted Frames:\n");
    while (current_pos < strlen(data)) {
        int actual_frame_len = 0;
        // Calculate the actual frame length, ensuring it doesn't exceed the data string
        if ((strlen(data) - current_pos) < frame_size) {
            actual_frame_len = strlen(data) - current_pos;
```

```

    } else {
        actual_frame_len = frame_size;
    }

    // Print the character count for the current frame
    printf("%d ", actual_frame_len);

    // Print the characters of the current frame
    for (k = 0; k < actual_frame_len; k++) {
        printf("%c", data[current_pos + k]);
    }
    printf("\n"); // New line for the next frame
    current_pos += actual_frame_len;
}
return 0;
}

```

OUTPUT:

1. Enter the data string: ACECOLLEGE
Enter the desired frame size: 4

The Total number of chars are :10
Transmitted Frames:
4 ACEC
4 OLLE
2 GE

2. Enter the data string: CSDDEPARTMENT
Enter the desired frame size: 5

The Total number of chars are :13
Transmitted Frames:
5 CSDDE
5 PARTM
3 ENT

1.2 BYTE STUFFING:

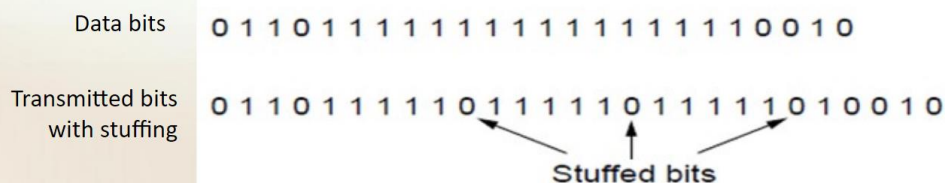
DESCRIPTION:

- In byte **stuffing** (or character stuffing), a special byte is added to the data section of the frame when there is a character with the same pattern as the flag.
- The data section is **stuffed** with an extra byte. This byte is usually called the escape character (ESC), which has a predefined bit pattern.
- Whenever the receiver encounters the ESC character, it removes it from the data section and treats **the next** character as data, not a delimiting flag.

Framing – Bit stuffing

Stuffing done at the bit level:

- Frame flag has six consecutive 1s (not shown)- called a bit pattern or flag byte
- On transmit, after five 1s in the data, a 0 is added
- On receive, a 0 after five 1s is deleted
- Flag bytes only occur at frame boundaries= not within the data



SOURCE CODE:

```
#include <stdio.h>
#include <string.h>
```

```
#define FLAG 'F' // Define Flag character
#define ESC 'E' // Define Escape character
```

```
int main() {
    char data[100];
    char stuffedData[200]; // Array to store stuffed data (larger to accommodate stuffed
    characters)
    int i, j = 0;

    printf("Enter the data string: ");
    scanf("%s", data);

    // Add starting Flag character
    stuffedData[j++] = FLAG;

    // Iterate through the data and perform stuffing
    for (i = 0; i < strlen(data); i++) {
        if (data[i] == FLAG || data[i] == ESC) {
```

```
        stuffedData[j++] = ESC; // Stuff an ESC character
    }
    stuffedData[j++] = data[i]; // Add the original data character
}

// Add ending Flag character
stuffedData[j++] = FLAG;
stuffedData[j] = '\0'; // Null-terminate the stuffed string

printf("Original Data: %s\n", data);
printf("Stuffed Data: %s\n ", stuffedData);

return 0;
}
```

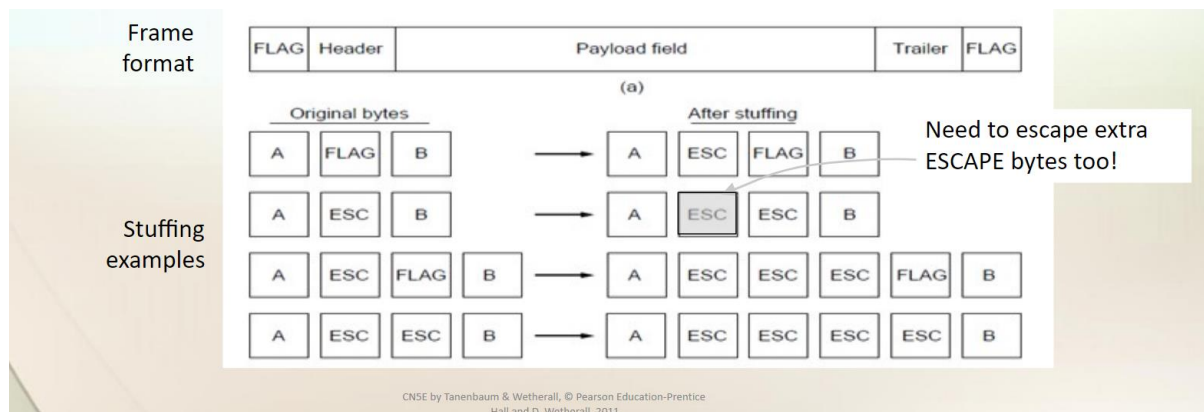
OUTPUT:

1. Enter the data string: ACE
Original Data: ACE
Stuffed Data: FACEEF
2. Enter the data string: CSDESC
Original Data: CSDESC
Stuffed Data: FCSDEESCF

1.3 BIT STUFFING

DESCRIPTION:

- To prevent *data* being interpreted as *control* information. For example, many frame-based protocols, such as X.25, signal the beginning and end of a frame with six consecutive 1 bits. Therefore, if the actual data being transmitted has six 1 bits in a row, a zero is inserted after the first 5 so that the data is not interpreted as a frame delimiter. Of course, on the receiving end, the stuffed bits must be discarded.
- For protocols that require a fixed-size frame, bits are sometimes inserted to make the frame size equal to this set size.
- For protocols that required a continuous stream of data, zero bits are sometimes inserted to ensure that the stream is not broken.



SOURCE CODE:

```
#include <stdio.h>
#include <string.h>
```

```
int main() {
    char data[100], stuffedData[200];
    int i, count = 0, j = 0;

    printf("Enter the data: ");
    scanf("%s", data);

    for(i = 0; i < strlen(data); i++) {
        if(data[i] == '1') {
            count++;
            stuffedData[j++] = data[i];
        } else {
            count = 0;
            stuffedData[j++] = data[i];
        }
    }
}
```

```
    if(count == 5) {  
        count = 0;  
        stuffedData[j++] = '0';  
    }  
}  
  
stuffedData[j] = '\0';  
  
printf("Data after bit stuffing: %s\n", stuffedData);  
  
return 0;  
}
```

OUTPUT:

1. Enter the data: 01111110
Data after bit stuffing: 011111010
2. Enter the data: 011111111111111101
Data after bit stuffing: 011111011111011111001

VIVA QUESTIONS FOR PROGRAMS-1

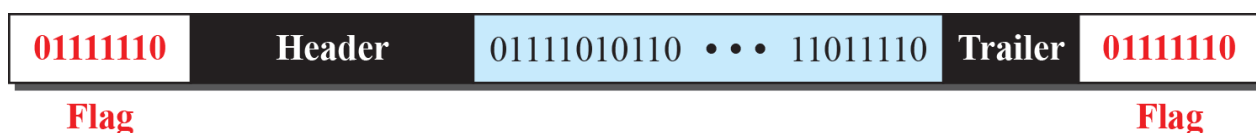
Framing

- Framing is a technique performed by the Data Link layer where data is transmitted as a stream of bits.
- There are two different framing techniques :
 - Fixed size framing
 - Variable size framing

Framing Approaches :

- Fixed length Approach
- Variable length Approach

A frame in a bit Stuffing



- Each frame begins and ends with a special bit pattern called a flag byte [01111110]. {Note this is 7E in hex}
- Whenever sender data link layer encounters five consecutive ones in the data stream, it automatically stuffs a 0 bit into the outgoing stream.
- When the receiver sees five consecutive incoming ones followed by a 0 bit, it automatically destuffs the 0 bit before sending the data to the network layer.

A frame in a Character Stuffing

- Each frame begins and ends ASCII characters (e.g. DLE STX and DLE ETX) referred to as character stuffing.
- The problem occurs when these character patterns occur within the “transparent” data.
- Solution: sender stuffs an extra DLE into the data stream just before each occurrence of an ‘accidental’ DLE in the data stream.
- The data link layer on the receiving end un-stuffs the DLE before giving the data to the network layer.

